

Boris Kafidov

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EDUCATION

University of Toronto

Honours B.Sc. in Computer Science & Mathematics, Focus in Artificial Intelligence

Toronto, ON

Expected Jun. 2028

EXPERIENCE

Amazon

Jr. Software Development Engineer (Internship)

Toronto, ON

Jun. 2026 – Present

- Shipped a TypeScript product-page component fix across three regions after tracing pre-order description errors to missing release-date context; preserved the request contract and added two regression tests to a 707-test suite
- Authored a component-migration design for book-series pages averaging 14.3M weekly visits across 12 marketplaces; led review with four engineers, time-boxed a cross-team feasibility check, and selected the API-backed path
- Shipped a three-region Ruby monitoring-as-code fix that reclassified a non-actionable self-recovering host event from Sev-2 to Sev-3 while preserving Sev-2 escalation for a genuine service failure; verified both paths in production

University of Toronto, Wheeler Lab

Software Engineer (Work Study)

Toronto, ON

Oct. 2025 – Mar. 2026

- Built simulated stage and camera backends for a Python/PySide6 Linux microscope-control application, enabling hardware-independent development; fixed a missed-signal race that caused false 10-second tiling waits
- Unified excitation hardware behind a shared controller interface; implemented serial and HID backends with automatic selection, readiness polling, timeouts, retries, and safe shutdown
- Instrumented researchers' long microscope runs with buffered JSONL profiling; batched dependency-ordered device updates and replaced sequential waits with one grouped Qt-signal wait plus timeout fallback

RESEARCH

A 13-vertex counterexample to e -log-concavity for chromatic quasisymmetric functions

Jul. 2026

Sole-authored arXiv preprint. Disproves Abreu–Nigro Conjecture 5.3; includes a public exact-arithmetic verifier

Dittert's conjecture in dimension 16 via a joint-deficit scaling lemma

Jul. 2026

Sole-authored arXiv preprint. Proves dimension 16 via matrix scaling and permanent bounds, settling all $n \geq 16$

PROJECTS

CreditCombo | Live App | JavaScript, Web Workers, Cloudflare Workers

Feb. 2026 – Mar. 2026

- Built and launched a browser-based optimizer that selects Canadian credit-card portfolios and category-level usage strategies by modeling net annual value across 151 cards, 51 rewards programs, and 22 spend subcategories
- Implemented deterministic, pruned combination search that prices rewards using estimated and minimum-guaranteed valuations, reroutes shared-cap overflow, and enforces locked-card and annual-fee constraints
- Kept the browser UI responsive by running combinatorial search in a Web Worker, discarding stale results via request IDs, and using a bounded cache; deployed through a Cloudflare Worker with route rewrites and CSP

APA Site Choice Transformer | Python, PyTorch, Hugging Face, GCP

Sep. 2025

- Co-led a five-person team to second place at the Toronto Bioinformatics Hackathon (\$500 award)
- Built and tuned a DNABERT-2-117M training pipeline for approximately 800k labeled sequence windows on GCP-hosted A100 GPUs using mixed precision and gradient accumulation
- Offloaded logits to CPU to prevent evaluation OOMs; implemented deterministic evaluation and optional chromosome holdouts; the team's model achieved 0.93 AUROC and 0.94 AUPRC

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Java, C++, C

Frameworks/Libraries: PyTorch, Hugging Face Transformers, PySide6, scikit-learn, NumPy, Pandas, Node.js

Platforms/Tools: Linux, Git, AWS CDK, CloudWatch, GCP, Cloudflare Workers, Bash

Areas: Machine Learning, Deep Learning, Algorithms & Data Structures, Concurrency, Performance & Reliability